

GENED 1104: SCIENCE & COOKING MIDTERM 2

Thursday, November 17, 2022, 12-1:15pm

Calculators permitted

Expectations

- **Read the directions carefully.**
- **For problems involving calculation, please show your work.**
- **Academic Integrity:**

Cheating, no matter how slight, is never acceptable.
It violates the standards of our community and disrupts learning.
A higher score is not worth compromising your integrity.

- I have not consulted anyone or any reference material in completing this exam.
- I will not discuss exam questions and solutions until the exam is handed back.
- I understand and agree to abide by this policy:

Name (Printed)

(Signature)

Section Time (circle):

M 9:45am M 12:45pm M 6:45pm T 3:45pm T 6:45pm W 12:45pm W 3:45pm W 6:45pm

R 9:45am R 3:45pm F 9:45am F 12:45pm F 3:45pm

TF (circle):

Julia Alex Heide Kate Jalilah Davide Lauren Audrey Claudia

Grading (to be completed by teaching staff)

Question	Score
1-5	/15
6-7	/10
8-11	/18
12-14	/19
15-20	/38
Total	/100

Multiple Choice Questions: (3 pts each)

- Which of the following food ingredients can be used as emulsifiers? Select all that apply.
 - Egg
 - Lemon juice
 - Mustard
 - Vinegar
- A “cappuccino” is a coffee beverage made with espresso and steamed, frothed milk. It is a.... (Select all that apply)
 - Foam
 - Colloidal dispersion
 - Emulsion
 - Crystal

- Perfectly tempered chocolate contains Phase V crystals. Which of the following are true? (Select all that apply).

- Heating to 50-55°C as a first step results in many nucleation sites
- Adding some tempered chocolate before heating up to 34°C allows for more Phase V crystals
- Phase V chocolate has a shiny surface
- All phases consist of sugar molecules with either double- or triple chain packing

Crystallization form	Melting Point
Phase I	18°C
Phase II	22°C
Phase III	26°C
Phase IV	29°C
Phase V	34°C
Phase VI	36°C

- You just made a delicious mac and cheese sauce. You put the still warm sauce in the fridge (4°C) overnight. The next morning you notice that the sauce has become more viscous compared to yesterday. Which of the following explains what has happened? (Select all that apply)
 - Proteins in the cheese sauce denatured and formed crosslinks in the cold temperature
 - As the sauce cools, less thermal energy makes molecules move past each other more slowly
 - As the sauce cools, less thermal energy causes droplets in the sauce to coalesce
 - The lower temperature allows the fats molecules in the sauce to form stronger hydrogen bonds
- You want to make a small serving of a sponge cake so you decide to halve all the ingredients in your usual recipe. Assuming you use the same 9” round pan to bake the cake in (i.e. the radius of the cake stays the same), how should you adjust the cooking time?
 - You don’t need to adjust, the cooking time is the same in both cases
 - Double the normal cooking time
 - Halve the normal cooking time
 - Use a quarter of the normal cooking time

Short Answer Questions:

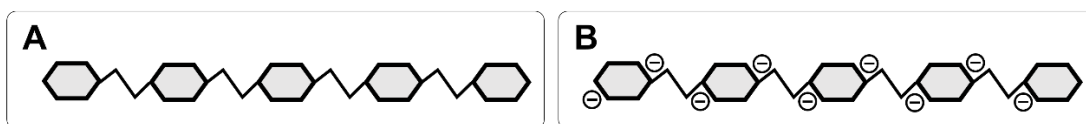
6. Crystal nucleation of cocoa butter has been found to increase upon addition of fats such as triolein. Do you expect the activation energy to increase or decrease when triolein is added to cocoa butter? Give one reason why. (5 points)

The activation energy should decrease when fat is added. [2 pts]

Could explain either with the equation of the week (N is inversely proportional to ΔG) or conceptually where lower activation energy means it is easier to create contacts to create a crystal lattice or contacts and nucleate. [3 pts for explanation]

Other explanations could be that the triolein allows it to undergo heterogeneous nucleation which is much faster.

7. Corn starch is an effective thickener. It is typically made up of long chains of neutral sugars [Figure A]. If instead these chains contained many negative charges [Figure B], would cornstarch be more or less effective at increasing viscosity? Explain your answer. (5 pts)



Less effective [2 pts]

Viscosity is a measure of a liquid's resistance to flow or, in other words, the ability of molecules in the liquid to flow past each other.

Starch granules normally leak polymers that stretch out and entangle together, therefore taking a long time to flow past each other and leading to a higher viscosity.

Negative charges on the starch would repel each other, allowing for them to flow over each other more easily and thus would decrease the viscosity.

[3 pts for explanation]

Celebratory Brunch

To celebrate finishing your last Science and Cooking midterm, you've decided to hold a big Sunday brunch. For your main dish you plan to make eggs benedict, which consists of two halves of an English muffin, each topped with Canadian bacon, a poached egg, and hollandaise sauce.

8. The hollandaise sauce recipe says to whisk egg yolks, then slowly add in butter until you get a creamy sauce. What method of thickening are you using here? Briefly explain how this method increases viscosity. (5 pts)

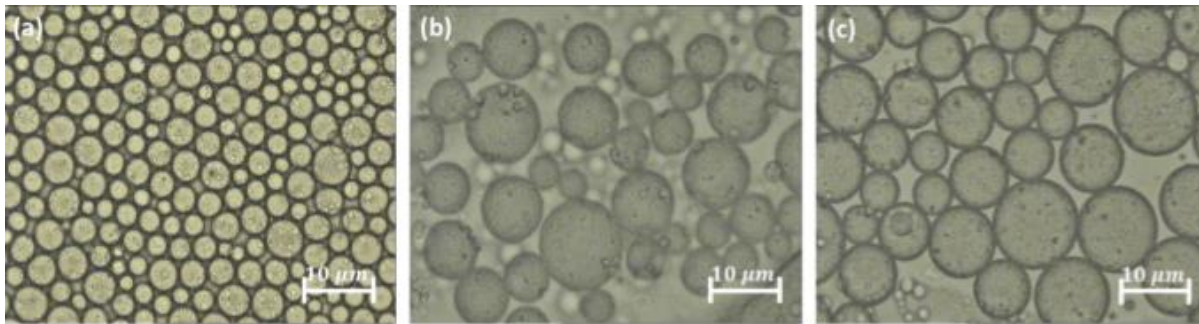
Forming emulsions [2 pts]: At sufficiently high volume fraction, droplets of an immiscible fluid act like very large particles to increase viscosity [3 pts]

9. You calculate that the volume fraction of your hollandaise sauce is 0.61. Would this sauce be considered a solid? Would it be considerably viscous? Explain each answer. (4 pts)

Not a solid--less than the critical vol fraction of 0.64. [2 pts]

Considerably viscous because it's over the 50-60% threshold. [2 pts]

You are curious if mixing the hollandaise sauce for different lengths of time will change the texture. You use a microscope to compare sauce samples which have been blended for a total of 1 minute, 3 minutes and 6 minutes. Based on the microscope images below, answer the following questions:



SLIDE A

SLIDE B

SLIDE C

10. Predict which slide corresponds to the hollandaise sauce blended for 1 minute, 3 minutes and 6 minutes total. Explain your prediction. (5 pts)

Blending creates shear stress which causes droplets to get smaller [2 pts]

Slide A – 6 min – droplets are smallest [1 pt]

Slide B – 3 min – droplets of intermediate size, some are smaller vs some are larger [1 pt]

Slide C – 1 min [1 pt]

11. Predict which sauce will have the highest elastic modulus. Explain your prediction. (4 pts)

Droplet size is inversely proportional to elasticity, so smallest droplet size = highest elasticity [2 pts]

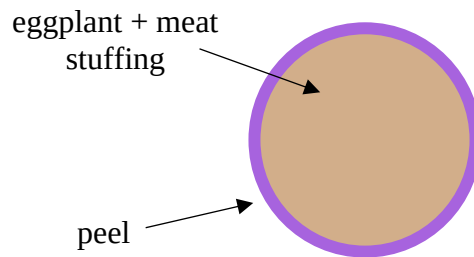
-> slide A (3 minute blended) [2 pts]

Stuffed Eggplant

Stuffed eggplant is a popular dish in many Mediterranean countries, and consists of an eggplant peel stuffed with a meat-and-eggplant mixture and baked in the oven at 200°C until golden brown. There are two possible preparation methods:

- A. Raw eggplant and meat are combined together and placed inside the eggplant peel – all ingredients are at room temperature. The stuffed eggplant is then baked in the oven until the center reaches 60°C (perfectly cooked).
- B. The eggplant and meat stuffing are pre-cooked together on the stove until they reach 60°C (perfectly cooked), and then placed inside the raw eggplant peel. Next, the stuffed eggplant is baked in the oven just until the outside is golden brown.

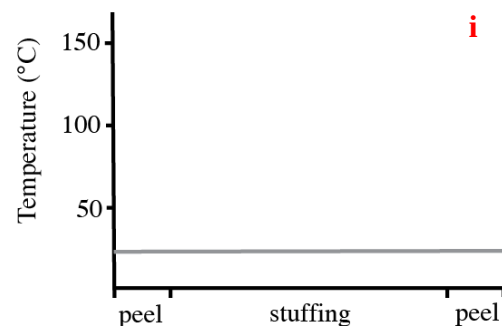
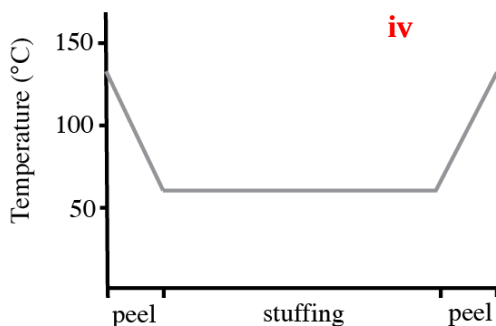
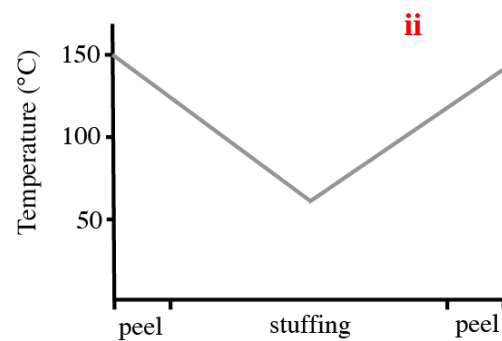
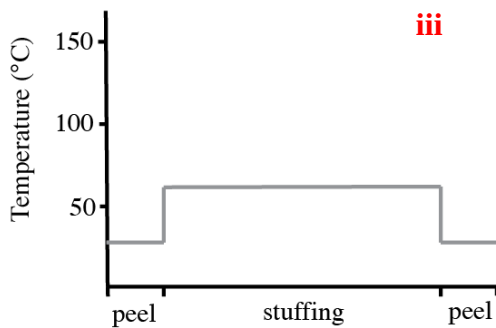
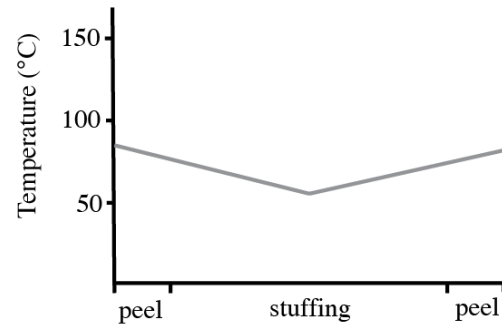
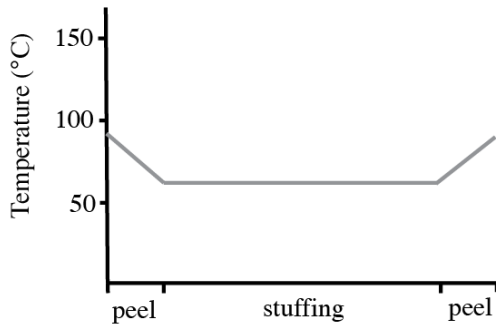
Below is a simplified cross section of the stuffed eggplant:



12. What process is turning the stuffed eggplant golden brown in the oven? What are the necessary components and conditions for this reaction? (5 pts)
- a. Maillard reaction leads to the brown color [2 pts]
 - b. Amino acids [1 pt] + carbs/sugars [1 pt] + high heat (over 120C) [1 pt]
13. What are two possible worries with preparation method A that would likely be avoided with method B? (6 pts)
- a. Since A is likely in the oven longer to fully cook the whole eggplant + stuffing, it is possible that (1) by the time the inside reaches temperature, the outside would be burned (since it is cooked at high heat), and (2) the outer parts of the inside might overcook while waiting for the very inside to become cooked
 - b. Also OK to say (3) A would take longer to cook overall b/c heating the mixture in a pan while mixing is faster than waiting for diffusion to reach the center

14. Of the temperature profiles below, label which graph corresponds to the cross section of the stuffed eggplant in the following scenarios: (8 pts) [2 pts per assignment]

- Preparation method A, right before being placed in the oven
- Preparation method A, after being baked just until the center is perfectly cooked
- Preparation method B, right before being placed in the oven
- Preparation method B, after being baked just until golden brown



Fruit Juice Meringues

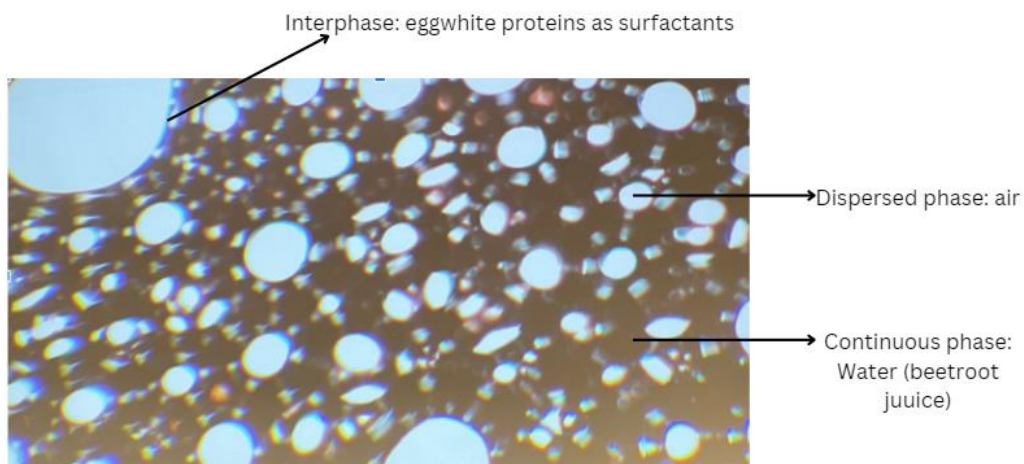
Inspired by Chef Eduard's beetroot meringue, you want to make fruit juice meringues as a dessert. You combine 350 g of fruit juice, 15 g of powdered egg whites, and 0.6 g of xanthan gum in a stand mixer, whipping for 4 minutes.

15. Xanthan gum is used in this recipe to help stabilize the meringue. What is one process that leads to instability in foams and explain how xanthan gum would help prevent it. (6 pts)

Foam instability is due to coalescence, drainage, ostwald's ripening... [2 pts]

Increases the viscosity of the continuous phase [2 pts] which prevents coalescence, sedimentation, drainage, etc. [2 pts] ... making the foam more stable.

16. Draw a diagram of how your meringue would look if you observed it under a microscope after you finished mixing. Indicate on the picture where each of the ingredients would be found (include air from the mixing). (8 pts)



Xanthan gum is dissolved/tangled into the continuous phase. [2 pts per ingredient]

17. Just as you are ready to bake the meringue, you notice that the meringue is “leaking” water. You decide to add lemon juice to the mixture and re-whip. Do you think this is a good idea? Explain why. (5 pts)

Yes, lemon juice will denature the proteins [2 pts]

This would expose the hydrophilic parts of the protein which can oriented to the dispersed phase and stabilize the air/water interface => STRONGER SURFACTANT, more stable foam.

Other possible answer: denatured proteins will tangle together, forming a network like the xanthan gum, increasing viscosity more and further stabilizing the foam

[3 pts explanation]

18. Assuming the average air bubble in your meringue is a sphere with a diameter of 25 μm ($1 \mu\text{m} = 10^{-6} \text{m}$), and there are 10^8 bubbles per mL ($1\text{mL} = 10^{-6} \text{m}^3$), what is the volume fraction of air incorporated into your meringue? (8 points)

V_{bubble} :

$$V_{\text{bubble}} = \frac{4}{3} * \pi r^3$$

$$V_{\text{bubble}} = \frac{4}{3} * 3.14 * (12.5 \times 10^{-6} \text{ m})^3 = 8177 \times 10^{-18} \text{ m}^3 = 8.1 \times 10^{-15} \text{ m}^3$$

In 1mL of foam:

$$V_{\text{dispersed (air)}} = V_{\text{bubble}} * 10^8 \text{ bubbles} = 8.1 \times 10^{-7} \text{ m}^3$$

$$\text{In 1mL: } V_{\text{total}} = 1\text{mL} = 10^{-6} \text{ m}^3$$

$$\text{Volume fraction: } \phi = V_{\text{dispersed}} / V_{\text{total}}$$

$$\phi = 8.1 \times 10^{-7} \text{ m}^3 / 10^{-6} \text{ m}^3 = 0.81$$

19. Compare the number you calculated in #18 with the maximum theoretical volume fraction for random packing of spheres. What is the maximum theoretical volume fraction? Given this, what are 2 explanations for how the bubbles are able to achieve the volume fraction in #18? (5 points)

Volume fraction: 0.64 for random packing of spheres. [1 pt]

- i. Higher volume fractions are possible because bubbles are (1) deformable [2 pts] and (2) polydisperse [2 pts]. The 64 % volume fraction is the max for spherical drops that are rigid and of the same size. Above 64 % drops will have flat contact areas where they are pressing/squished against one another. Also, bubbles tend to be a mix of different sizes – so small bubbles can fit in the space between larger bubbles.

20. If your meringue has an elastic modulus of 500 Pa, calculate the surface tension of the air bubbles. To account for other ingredients assume a total volume fraction of 0.90. (6 pts)

$$\begin{aligned} E &= \sigma / R (\varphi - \varphi_{\text{critical}}) \\ 500 \text{ Pa} &= \sigma / 12.5 \times 10^{-6} * (0.90 - 0.64) \\ \sigma &= 0.024 \text{ N/m} \end{aligned}$$

SCRATCH PAPER

Equation Sheet

Unit conversions				
$N = \text{kg} \cdot \text{m/s}^2$	$J = \text{N} \cdot \text{m}$	$\text{Pa} = \text{N/m}^2 = \text{J/m}^3$	$1 \text{ L} = 1 \cdot 10^{-3} \text{ m}^3$	$1 \text{ ml} = 1 \text{ cm}^3$
Kelvin = Celsius+273	Fahrenheit = (Celsius*9/5)+32	Area circle = πr^2	Volume sphere = $(4/3)\pi r^3$	Surface area sphere = $4\pi r^2$
Constants				
Room temp = 23°C	1Cal = 1000 cal = 4.18 kJ	Gravitational acceleration (g=-9.8 m/s ²)	Avogadro's number (6.022·10 ²³ molecules/mol)	
Boltzmann constant $k_B = (1.38 \cdot 10^{-23} \text{ J/K})$	Specific heat water 4.19 J/gK	Latent heat of vaporization for water 2260 J/g	Latent heat of fusion for water 334 J/g	
1 mol = 6.022 x 10 ²³ molecules	gas constant = 22.4L/mol	D _{heat} water 1.4·10 ⁻³ cm ² /sec	D _{Ca} 8.0·10 ⁻¹⁰ m ² /sec	

Equation	Description	Units
$L = \sqrt{4Dt}$	L Distance of diffusion	m
	D Diffusion coefficient	m ² /s
	t Time to diffuse	s
$\phi = \frac{V_{dispersed}}{V_{total}}$	ϕ Volume fraction	-
	$V_{dispersed}$ Volume of dispersed phase	L
	V_{total} Total volume	L
$E = \frac{S}{R}(f - f_c) \quad f > f_c$	E Elastic modulus	Pa
	σ Surface tension	N/m
	R Droplet radius	m
	ϕ Dispersed phase volume fraction	-
	ϕ_c Critical volume fraction (0.64 for spheres)	-
$N \sim e^{-\frac{DG}{k_B T}}$	N Number of nuclei	-
	ΔG Activation energy	J
	k_B Boltzmann constant	J/K
	T Temperature	K

Equation		Description	Units
$\text{pH} = -\log_{10}[\text{H}^+]$	$[\text{H}^+]$	Concentration of protons (also known as H^+ ions) in solution	mol/L
	pH	A measure of acidity	none
$Q = mc_p \Delta T$	Q	Heat	J
	m	Mass	g
	c_p	Specific heat capacity	J/g K or J/g °C
	ΔT	Change in temperature	K or °C
$Q = mL_v$ $Q = mL_f$	Q	Heat	J
	m	Mass	g
	L_v	Latent heat of vaporization	J/g
	L_f	Latent heat of fusion	J/g
$U = Ck_B T$	U	Interaction energy for liquid to gas transitions	J
	C	Constant (1.5 for water)	none
	k_B	Boltzmann constant	J/K
	T	Temperature	K
$E = \frac{F / A}{\Delta L / L_0}$	E	Elastic modulus	Pa
	F	Force	N
	A	Area over which force is applied	m ²
	ΔL	Change in length	m
	L_0	Initial length	m
$E = \frac{U_{\text{int}}}{l^3} = \frac{k_B T}{l^3}$	E	Elastic modulus	Pa
	k_B	Boltzmann constant	J/K
	T	Temperature	K
	l	Average distance between crosslinks	m
$N(t) = N_0 e^{kt}$ where $k = \frac{\ln 2}{t}$	$N(t)$	Number of cells at time t	microbes
	N_0	Number of cells at time 0	microbes
	k	Growth or death rate of cells	s ⁻¹
	t	Time elapsed	s
	τ	Doubling time	s